10.1.1 Mathematical formulation of a linear programming problem:

The procedure for mathematical formulation of a linear programming problem consists of the following steps.

(i) Identify the decision variables.

(ii) Identify the objective function to be maximized or minimized and express it as a linear function of decision variables.

(iii) Identify the set of constraint conditions and express them as linear inequalities or equations in terms of the decision variables.



Linear Programming Problem is used in determining shortest routes for travelling salesman.

Example 10.1

A furniture dealer deals only two items viz., tables and chairs. He has to invest ₹10,000/- and a space to store atmost 60 pieces. A table cost him ₹ 500/– and a chair ₹ 200/–. He can sell all the items that he buys. He is getting a profit of ₹ 50 per table and ₹ 15 per chair. Formulate this problem as an LPP, so as to maximize the profit.

Solution:

(i) Variables:

Let x_1 and x_2 denote the number of tables and chairs respectively.

(ii) Objective function:

Profit on x_1 tables = $50\,x_1$

Profit on x_2 chairs = $15\,x_2$

Total profit = $50\,x_1+15x_2$

Let $Z = 50\,x_1+15\,x_2$, which is the objective function.

Since the total profit is to be maximized, we have to maximize $Z=50x_1+15x_2$

(iii) Constraints:

The dealer has a space to store atmost 60 pieces

$x_1+x_2 \le 60$

The cost of x_1 tables = ₹ $500\,x_1$

The cost of x_2 tables = ₹ $200\,x_2$

Total cost = $500\,x_1 + 200\,x_2$,which cannot be more than 10000

$500\,x_1 + 200\,x_2 \le 10000$

$5\,x_1 + 2x_2 \le 100$

(iv) Non-negative restrictions:

Since the number of tables and chairs cannot be negative, we have $x_1 \ge 0, x_2 \ge 0$

Thus, the mathematical formulation of the LPP is

Maximize $Z=50\,x_1+15x_2$

Subject to the constrains

$x_1+x_2 \le 60$

$5x_1+2x_2 \le 100$

$x_1, x_2 \ge 0$

Example 10.2

A company is producing three products P_1, P_2 and P_3, with profit contribution of ₹ 20, ₹ 25 and ₹ 15 per unit respectively. The resource requirements per unit of each of the products and total availability are given below.

Product	P_1	P_2	P_3	Total availability
Man hours/unit	6	3	12	200
Machine hours/unit	2	5	4	350
Material/unit	1kg	2kg	1kg	100kg

Formulate the above as a linear programming model.

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